MSc Project Report

Exponential Sums and Weil Bounds: A Survey of Elementary Methods and Decoding Applications

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Abstract

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Preface

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	**Meaning**
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^\infty e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^\top C_f X$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i [d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	Meaning
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha\in\mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha\in\mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha\in\mathbb{F}_q} \psi(f(\alpha))\,\chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha\in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha\in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha\in E} \psi^{(k)}(f(\alpha))\,\chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h\in\Phi} \xi(h)\,\mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h\in\Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k\cdot\deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$
Hasse Derivatives	
$F^{(\ell)}(X)$	ℓ-th Hasse derivative of $F(X)$: coefficient of Z^ℓ in $F(X + Z)$
$\mathrm{mult}(F, \alpha)$	Multiplicity of vanishing of F at α: largest r with $(X - \alpha)^r \mid F(X)$
Pseudopolynomials	
$\Lambda(X)$	The polynomial $X^q - X$ over $\mathbb{F}_q$
$F_{\langle\ell\rangle}(X)$	ℓ-th pseudoderivative of F: unique polynomial of degree $< q$ agreeing with $F^{(\ell)}$ on $\mathbb{F}_q$
$\mathrm{pdeg}(F)$	Pseudodegree of F: $\max_{\ell\geq 0} \deg(F_{\langle\ell\rangle})$
$F_i(X)$	Coefficient polynomials in the base-Λ expansion $F = \sum_i F_i \Lambda^i$, with $\deg(F_i) < q$
Decoding from Character Evaluations	
$r : \mathbb{F}_q \to \{0, \pm 1\}$	Received (noisy) word; approximates $\psi \circ g$ or $\mathrm{Tr} \circ g$

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Symbol	**Meaning**
e	Number of errors: $\lvert\{\alpha \in \mathbb{F}_q : r(\alpha) \neq (\psi \circ g)(\alpha)\}\rvert$
S	Error set: $\{\alpha \in \mathbb{F}_q : r(\alpha) \neq (\psi \circ g)(\alpha)\}$
$Z_S(X)$	Error locator polynomial: $\prod_{\alpha \in S}(X - \alpha)$
$\Delta(r_1, r_2)$	Hamming distance: $\lvert\{\alpha \in \mathbb{F}_q : r_1(\alpha) \neq r_2(\alpha)\}\rvert$
$G(X)$	Auxiliary polynomial $g(X)^{(q-1)/2+M}$ encoding the character values
$E(X)$	High-multiplicity error-locating pseudopolynomial; multiple of Z_S^M
y_F	Codeword associated to a k-pseudopolynomial F: $\alpha \mapsto (F^{(0)}(\alpha), \ldots, F^{(M-1)}(\alpha))$

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